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| **Fiscal Year:** | FY 2022 | **Task Last Updated:** | FY 03/02/2022 |
| **PI Name:** | Koochesfahani, Manoochehr Ph.D. | | |
| **Project Title:** | ZBOT-2 Thermometry Diagnostics Development | | |
| **Division Name:** | Physical Sciences | | |
| **Program/Discipline:** | | | |
| **Program/Discipline--Element/Subdiscipline:** | FLUID PHYSICS--Fluid physics | | |
| **Joint Agency Name:** | | **TechPort:** | No |
| **Human Research Program Elements:** | None | | |
| **Human Research Program Risks:** | None | | |
| **Space Biology Element:** | None | | |
| **Space Biology Cross-Element Discipline:** | None | | |
| **Space Biology Special Category:** | None | | |
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| **Zip Code:** | 48824 | **Congressional District:** | 8 |
| **Comments:** | | | |
| **Project Type:** | Ground | **Solicitation / Funding Source:** | Physical Sciences Unsolicited |
| **Start Date:** | 05/01/2019 | **End Date:** | 04/30/2021 |
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| **No. of PhD Candidates:** | | **No. of Master' Degrees:** | |
| **No. of Master's Candidates:** | | **No. of Bachelor's Degrees:** | |
| **No. of Bachelor's Candidates:** | | **Monitoring Center:** | NASA GRC |
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| **Flight Program:** | | | |
| **Flight Assignment:** | | | |
| **Key Personnel Changes/Previous PI:** | | | |
| **COI Name (Institution):** | | | |
| **Grant/Contract No.:** | 80NSSC19M0128 | | |
| **Performance Goal No.:** | | | |
| **Performance Goal Text:** | | | |
| **Task Description:** | NOTE: Follow on project to "ZBOT Thermometry and Velocimetry Diagnostics Development," grant NNX16AD52A, with same Principal Investigator, Dr. Manoochehr Koochesfahani.<br>In an effort that was launched several years ago under NASA support (Grant No. NNX16AD52A), we demonstrated the feasibility of non-intrusive molecular based velocity and temperature measurement methods suitable for the Zero Boil-Off Tank (ZBOT) microgravity experiment. Molecular tagging velocimetry (MTV) adds complementary capability to the Digital Particle Image Velocimetry (DPIV) already utilized in the ZBOT experiment, whereas Quantum Dot Thermometry (QDT) using laser induced fluorescence (LIF) of solubilized quantum dots (QD) enables whole-field thermometry, a capability that did not exist previously.<br><br>Our original development of QDT relied on two-color ratiometric LIF thermometry using QDs solubilized in PnP and a pair of aligned monochrome cameras. The temperature measurement uncertainty was found to depend on several | | |

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| **Task Description:** | interconnected parameters, e.g., degree of misalignment between the two detectors, signal to noise ratio/dynamic range of the detector, and the spatial resolution of imaging (and therefore the size of field of view and pixel density of detector).<br><br>The objective of the current research effort is to capitalize on our previous progress in QDT diagnostics and more fully quantify the various interconnected parameters, and the tradeoffs among them, which influence the ability to make accurate temperature measurements. This type of information is needed to optimize the design of experiments for accurate quantitative measurements in the actual ZBOT test section on the International Space Station (ISS). As part of this study, we will also consider the feasibility of utilizing a single color camera, in place of two aligned monochrome cameras, due to concerns that the space launch process might adversely affect the two-camera alignment. |
| **Rationale for HRP Directed Research:** | |
| **Research Impact/Earth Benefits:** | The ZBOT project directly aligns with NASA exploration needs for Cryogenic Fluid Management of future propellant and life support storage tanks, and the ZBO tank pressure control is an important technology achievement that enables human missions to Moon, Mars, and other planets. |
| **Task Progress:** | In an effort that was launched several years ago under NASA support (Grant No. NNX16AD52A), we demonstrated the feasibility of non-intrusive molecular based velocity and temperature measurement methods suitable for the ZBOT microgravity experiment. Molecular tagging velocimetry (MTV) in the ZBOT-2 investigation has added complementary capability to the Digital Particle Image Velocimetry (DPIV) already utilized in the ZBOT experiment, whereas Quantum Dot Thermometry (QDT) using laser induced fluorescence (LIF) of solubilized quantum dots (QD) enables whole-field thermometry (a capability that did not exist previously).<br>Our original development of QDT relied on two-color ratiometric LIF thermometry, using QDs solubilized in Perfluoro-n-Pentane (PnP) and a pair of aligned monochrome cameras. The temperature measurement uncertainty was found to depend on several interconnected parameters – e.g., degree of misalignment between the two detectors; signal to noise ratio/dynamic range of the detector; and the spatial resolution of imaging (and therefore, the size of field of view and pixel density of the detector).<br><br>The objective of the ZBOT-2 research effort was to capitalize on our previous progress in QDT diagnostics and more fully quantify the various interconnected parameters – and the tradeoffs among them – that influence the ability to make accurate temperature measurements. This type of information is needed to optimize the design of experiments for accurate quantitative measurements in the actual ZBOT test section on the International Space Station (ISS). As part of this study, we also considered the feasibility of utilizing a single color camera, in place of two aligned monochrome cameras, due to concerns that the space launch process might adversely affect the two-camera alignment. The status of our progress at the conclusion of this two-year effort is described below.<br><br>1. Description of New Cameras and Temperature Response For this work, we considered high performance 16-bit PCO.panda bi (back-illuminated) black and white (B/W) cameras, as well as color cameras. A prediction was made of ratiometric temperature response of two filter sets. We noted that the Bayer color spectrum is not optimum for obtaining the best ratiometric temperature response from the emission spectra of the particular QDs we used. The predicted color camera temperature response was 2.5 times less than the B/W camera pair fitted with the optimized filter set. Actual measurements of the temperature response agreed very well with the predictions.<br><br>2. Proof of Concept Experiment with the Color Camera Even though the ratiometric temperature response of the color camera was found to be much lower than the B/W camera pair, we decided to carry out a preliminary proof-of-concept experiment in the ZBOT breadboard setup to get an early assessment of the color camera performance. The ambient tank fluid was initially set at room temperature of about 22.5°C. A cold jet, with a temperature of about 14°C, entered the tank from the bottom, pointing upwards with a speed estimated at ~ 2.2 cm/seconds. The center plane of the test section was illuminated with a 200 milliwatt, 405 nanometer laser sheet; the sheet was formed into a nearly uniform intensity distribution across a 9 cm height using a Powell lens. The central 4.6 cm × 4.6 cm region of the laser sheet was imaged onto the 2048 × 2048 pixel color camera, resulting in an image pixel resolution of 22.4 micrometers/pixel. The camera acquired images at a rate of 5 frames/second with an exposure time of 150 milliseconds.<br><br>3. QDT Performance Curves for the B/W Camera Pair and the Color Camera As mentioned, the temperature measurement uncertainty depends on a series of interconnected parameters. These parameters include: • the signal to noise ratio/dynamic range of the detector; • the spatial resolution of imaging (and therefore the size of field of view and pixel density of detector); • the imperfect alignment between detectors in the case of two B/W cameras; • the accuracy of intensity estimation at the missing pixels in the case of the color camera.<br><br>The tradeoff among these interconnected parameters, and the resulting temperature measurement uncertainty are specific to each experiment and the detection strategy utilized.<br><br>The discussion above led to our effort to establish the QDT performance curves for both detection strategies (i.e., single color camera and B/W camera pair), in terms of the expected instantaneous temperature measurement uncertainty versus the camera signal level and different spatial averaging domains (representing different spatial resolutions of the measured temperature field). The results will provide a master data set that help optimize the selection of experimental parameters for meeting the science requirements in the ZBOT experiments on the ISS.<br><br>4. New Challenges The second year of this research effort faced two major challenges. The COVID-19 pandemic led to the shutdown of our research laboratory for about 4.5 months, resulting in a significant delay in experiments. Instead, we focused on the analysis of large-eddy simulation (LES) and Reynolds-averaged Navier-Stokes (RANS) temperature field data sets from Dr. Kassemi's Computational Fluid Dynamics (CFD) group. We assessed the ability to discern the important structures in the temperature field data under the influence of temporal and spatial averaging of the data, and the expected temperature uncertainty of the measurement. The information gained from this analysis proved very instructive to the investigative team and to the CFD group in formulating the relevant science requirements.<br><br>The second major challenge was that the QDs we had been using for several years were no longer sold by the supplier. Therefore, the majority of the last six months of the project was focused on identifying alternate sources of QDs that can be solubilized/dispersed in PnP. There has been good progress in identifying PnP-solubilized QDs that are suitable for QDT. At the conclusion of the current two-year effort, we are pleased to report that initial measurements indicate our |

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| | first batch of QDs from NanoOptical Materials (NOM) is a viable replacement for the original QDs. Further analysis of the new QD and optimizing its performance are ongoing. |
| **Bibliography Type:** | Description: (Last Updated: 03/02/2026) |
| **Abstracts for Journals and Proceedings** | Koochesfahani M, Olson D. "Influence of detection strategy on the performance of whole-field planar optical diagnostics for thermometry in the ZBOT experiment." 35th Annual Meeting of the American Society for Gravitational and Space Research, Denver, CO, November 20-23, 2019.<br>Abstracts. 35th Annual Meeting of the American Society for Gravitational and Space Research, Denver, CO, November 20-23, 2019. , Nov-2019 |
| **Abstracts for Journals and Proceedings** | Olson D, Koochesfahani M. "Performance estimates of ratiometric quantum dot thermometry for the NASA ZBOT experiment." 73rd Annual Meeting of the APS Division of Fluid Dynamics, Virtual, November 22–24, 2020.<br>Bulletin of the American Physical Society. 2020 Nov;65(13):T09.00011.<br>https://meetings.aps.org/Meeting/DFD20/Session/T09.11 , Nov-2020 |
| **NASA Technical Documents** | Olson D, Blanchard G, Pouya S, Koochesfahani M. "Development of Quantum Dot Thermometry (QDT) for the ZBOT Experiment." Cleveland, Ohio : NASA Glenn Research Center, 2020. NASA/ZBOT-NC Systems Requirements Review. , Jun-2020 |